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Architecture

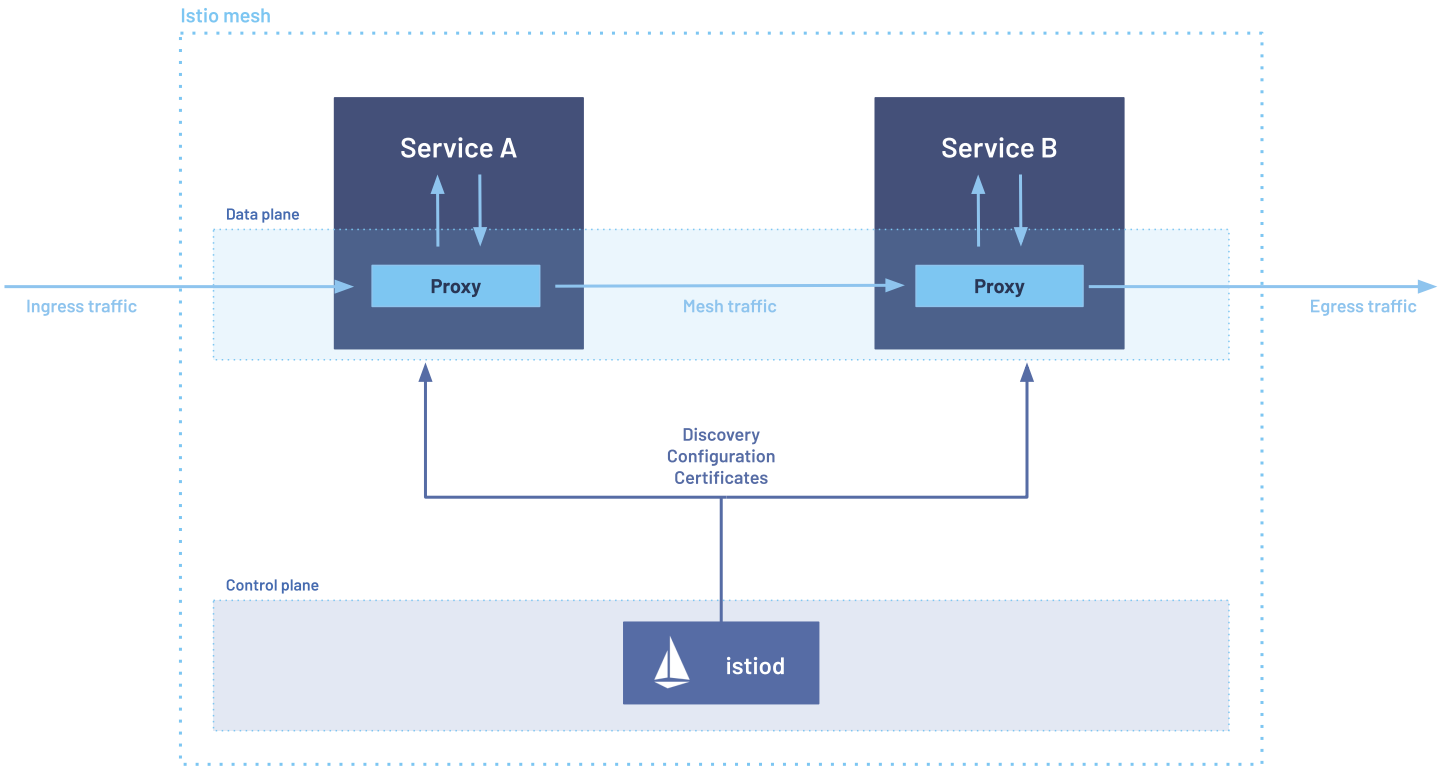
3 minute read

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An Istio service mesh is logically split into a **data plane** and a **control plane**.

- The **data plane** is composed of a set of intelligent proxies (Envoy[4]) deployed as sidecars. These proxies mediate and control all network communication between microservices. They also collect and report telemetry on all mesh traffic.
- The **control plane** manages and configures the proxies to route traffic.

The following diagram shows the different components that make up each plane:



Istio architecture in sidecar mode

Components

The following sections provide a brief overview of each of Istio's core components.

Envoy

Istio uses an extended version of the Envoy[4] proxy. Envoy is a high-performance proxy developed in C++ to mediate all inbound and outbound traffic for all services in the service mesh. Envoy proxies are the only Istio components that interact with data plane traffic.

Envoy proxies are deployed as sidecars to services, logically augmenting the services with Envoy's many built-in features, for example:

- Dynamic service discovery
- Load balancing
- TLS termination
- HTTP/2 and gRPC proxies
- Circuit breakers
- Health checks
- Staged rollouts with %-based traffic split
- Fault injection
- Rich metrics

This sidecar deployment allows Istio to enforce policy decisions and extract rich telemetry which can be sent to monitoring systems to provide information about the behavior of the entire mesh.

The sidecar proxy model also allows you to add Istio capabilities to an existing deployment without requiring you to rearchitect or rewrite code.

Some of the Istio features and tasks enabled by Envoy proxies include:

- Traffic control features: enforce fine-grained traffic control with rich routing rules for HTTP, gRPC, WebSocket, and TCP traffic.
- Network resiliency features: setup retries, failovers, circuit breakers, and fault injection.
- Security and authentication features: enforce security policies and enforce access control and rate limiting defined through the configuration API.
- Pluggable extensions model based on WebAssembly that allows for custom policy enforcement and telemetry generation for mesh traffic.

Istiod

Istiod provides service discovery, configuration and certificate management.

Istiod converts high level routing rules that control traffic behavior into Envoy-specific configurations, and propagates them to the sidecars at runtime. It abstracts platform-specific service discovery mechanisms and synthesizes them into a standard format that any sidecar conforming with the Envoy API[5] can consume.

Istio can support discovery for multiple environments such as Kubernetes or VMs.

You can use Istio's Traffic Management API to instruct Istiod to refine the Envoy configuration to exercise more granular control over the traffic in your service mesh.

Istiod security[6] enables strong service-to-service and end-user authentication with built-in identity and credential management. You can use Istio to upgrade unencrypted traffic in the service mesh. Using Istio, operators can enforce policies based on service identity rather than on relatively unstable layer 3 or layer 4 network identifiers. Additionally, you can use Istio's authorization feature to control who can access your services.

Istiod acts as a Certificate Authority (CA) and generates certificates to allow secure mTLS communication in the data plane.

Links

1. https://istio.io/latest/docs/
2. https://istio.io/latest/docs/ops/
3. https://istio.io/latest/docs/ops/deployment/
4. https://www.envoyproxy.io/
5. https://www.envoyproxy.io/docs/envoy/latest/api/api
6. https://istio.io/latest/docs/concepts/security/